

CALCULATING THEORETICAL PROBABILITY



LESSON INTRODUCTION

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

LEARNING TARGETS

- I can calculate the theoretical probability of the outcomes in a repeated experiment.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.1 Determine the sample space for a simple experiment.

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

WORKING TOWARD

MA.912.DP.4.1 Describe events as subsets of a sample space using characteristics, or categories, of the outcomes, or as unions, intersections or complements of other events.

MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.

ESSENTIAL QUESTIONS

- What is theoretical probability?
- How can I use the sample space to calculate theoretical probability?

LESSON NARRATIVE

In this lesson, students build on what they have learned about determining the sample space and sample space size to calculating theoretical probability of repeated experiments. Students will represent sample spaces using outcome tables and/or tree diagrams to build a ratio for the probability. The primary learning outcome in this lesson focuses on identifying the theoretical probability of an event in a repeated experiment.

COMPONENT SUMMARY

14.3.1 WARM-UP (~5 MIN.)

Solve proportions

14.3.3 GUIDED PRACTICE (10-15 MIN.)

Students determine sample spaces and calculate theoretical probability in new contexts

14.3.5 WRAP-UP (~5 MIN.)

Self-reflection on the lesson

14.3.2 GUIDED INSTRUCTION (10 MIN.)

Determine theoretical probability of a repeated experiment

14.3.4 YOUR TURN! (5-10 MIN.)

Independent practice calculating theoretical probability

RESOURCES

GLOSSARY

Key Terms:

- Theoretical Probability

Additional Terms:

- N/A

MATERIALS

- (optional) bag of marbles for 14.3.3 Guided Practice

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may misinterpret the events “RB” and “BR” as the same result.
- Students may miscalculate theoretical probability if they have an inaccurate sample space by not using outcome tables and/or tree diagrams.

THINGS TO CONSIDER

- Theoretical probability will regularly be compared with experimental probability starting in Lesson 4, therefore consider prompting students to think about experimental probability in these problems as well.

LITERACY AND LANGUAGE ROUTINES

Co-Craft Questions and Problems

14.3.2 Guided Instruction includes the game of rolling two fair six-sided dice. Since students are familiar with this game from previous lessons, they have a good understanding of the sample space already. Use this as an opportunity for students to explore the probabilities of other events by engaging in this language routine.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students apply the definition of theoretical probability to real-world contexts such as rolling dice, picking marbles, spinning spinners, etc. in this lesson. Consider having students also list some additional equally likely experiments that they can challenge each other to calculate the theoretical probability of.

DIFFERENTIATE INSTRUCTION

The contexts of these problems could be represented physically to help kinesthetic learners create the sample space. Although students should be able to create the sample space of the two dice since this has been repeated in preceding lessons, consider having marbles for the 14.3.3 Guided Practice since this is a new context to them.

Additional differentiation opportunities are denoted throughout the lesson guide.

14.3.1 WARM-UP: SOLVING PROPORTIONS

DIFFERENTIATE INSTRUCTION

Use observations from the Warm-Up to determine which students may need additional support as they work with probabilities in different forms in the upcoming lessons.

TEACHER MOVES

Take Turns

Consider having students engage in a Take Turns routine as they discuss and compare their answers during question 2. Have students take turns sharing their strategies for solving each proportion and explaining which one they found easiest to solve.



14.3.1 Warm-Up: Solving Proportions

1. Solve the proportions for the given variable.

a. $\frac{n}{54} = \frac{20}{100}$ $n = 10.8$

b. $\frac{x}{6} = 0.5$ $x = 3$

c. $20\% = \frac{6}{a}$ $a = 30$



14.3.2 Guided Instruction: Determining Probability

Probability can be used to better understand random processes. One way to analyze a random process is by calculating the theoretical probability of the events in a sample space.



Theoretical probability is a number between 0 and 1 that represents the likelihood of an event in a theoretical model based on a sample space. If all outcomes in the sample space are equally likely, then the theoretical probability of an event is the ratio of the number of outcomes in the event to the number of outcomes in the sample space.

1. Samaria plays a game that involves rolling two fair six-sided dice.
- a. Determine the outcomes in the sample space for the game.

		Die 2					
		1	2	3	4	5	6
Die 1	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

14.3.2 GUIDED INSTRUCTION: DETERMINING PROBABILITY

QUESTIONING

Consider having students construct an argument for the most common sum to roll when playing this game. After the students have made a written argument, have them find a group of other students that agree with them.

Then, a spokesperson from each group provides their reasoning to the class using mathematical language and reasoning.

14.3.2 GUIDED INSTRUCTION: DETERMINING PROBABILITY (CONTINUED)

TEACHER MOVES

The subsequent lessons in this unit include students performing experiments to evaluate the accuracy of the theoretical probabilities. Prompt students to begin thinking about the differences between theoretical and experimental probability.

- How many times do you think you would roll before rolling a sum of 9 or more?
- If you actually rolled 18 times, how many of those do you predict would result in a sum greater than or equal to 9?

QUESTIONING

Co-Craft Questions and Problems

Consider prompting the students to work in pairs to determine probability using the sample space from this game. Prompt the pairs to use mathematical language when crafting their questions or problems and develop their answers together orally.

- b. Samaria only wins if she rolls a sum of 9 or more. How many of the outcomes will result in a win?
10
- c. Since each outcome in the sample space is equally likely, a ratio can be created to compare the outcomes with a sum greater than or equal to 9 to the total number of outcomes.

$$P(\text{Sum} \geq 9) =$$

$$\frac{\text{the number of ways to get a sum} \geq 9}{\text{the total number of outcomes in the sample space}} = \frac{10}{36}$$

This ratio represents the probability of Samaria winning.

- d. Write the probability of winning as a fraction, decimal rounded to the nearest hundredth, and a percentage rounded to the nearest tenth.

Fraction: $\frac{5}{18}$

Decimal: **0.28**

Percentage: **27.8%**

To calculate a theoretical probability, think through the random process and use the sample space to determine the probability of certain events in the sample space.



14.3.3 Guided Practice: Calculating Theoretical Probability

- Ashley places 3 red marbles, 3 blue marbles, 3 yellow marbles, and 3 orange marbles in a bag. A representation of the marbles is shown.

 - What are the possible outcomes of randomly choosing one marble?
 $\{R,B,Y,O\}$
 - How many outcomes are in the sample space when choosing two marbles, with replacement?
16
 - Write the sample space for choosing two marbles from the bag, with replacement.
 $\{RR, RB, RY, RO, BR, BB, BY, BO, YR, YB, YY, YO, OR, OB, OY, OO\}$
 - Are all outcomes equally likely when choosing two marbles, with replacement? Explain your reasoning.
Answers will vary. Yes, because all of the colors have 3 marbles each.
 - Let Event A = two marbles of the same color are chosen. How many outcomes are in Event A ?
4
 - What is the probability of choosing two marbles of the same color, $P(A)$

Fraction	Decimal	Percentage
$\frac{1}{4}$	0.25	25%

14.3.3 GUIDED PRACTICE: CALCULATING THEORETICAL PROBABILITY

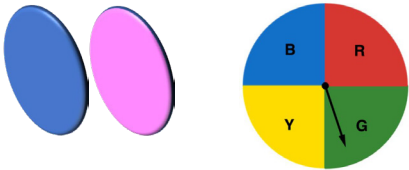
DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first part of the lesson to dictate how “guided” this Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

TEACHER MOVES

Students may misinterpret the events “RB” and “BR” as the same result. Consider prompting students to create an outcome table or tree diagram to see both events. It may also be helpful to have a kinesthetic approach to this by providing the students with marbles or other physical representations of this bag of marbles, such as colored slips of paper.

- A fair coin with one pink side and one blue side and a fair spinner are shown.



- If the coin is tossed and the spinner is spun, how many outcomes are in the sample space?
8
- Write the sample space.
 $\{BB, BR, BY, BG, PB, PR, PY, PG\}$
- What is the probability that both land on blue?
 $\frac{1}{8}$
- If the coin is tossed eight times, how many outcomes are in the sample space?
 2^8 or 256
- What is the probability that it lands on pink all eight times?
 $\frac{1}{256}$

TEACHER MOVES

Consider having students explain to their partner verbally how they calculated the probability. Prompt them to use mathematical language and refer back to the definition above. Ask guiding questions to facilitate the discussion like:

- What are other ways that you can represent this probability?
- What are other possible outcomes that would be more likely? Less likely? Equally likely?

14.3.4 YOUR TURN!

TEACHER MOVES

As students get more comfortable with probability, they often stop representing the sample space as outcome tables or tree diagrams. This often results in inaccurate sample spaces, which lead to inaccurate probabilities. Consider prompting students to reflect on the previous practice that was completed to see how an incorrect sample space could affect the probability calculation.

QUESTIONING

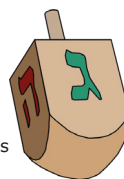
Although students will not be asked to solve probability without replacement in this unit, consider challenging early finishers with the following questions.

- How many outcomes are in the sample space when choosing two marbles and not using replacement?
- How does this differ from using replacement?



14.3.4 Your Turn!

1. A dreidel is a four-sided spinning top, played during the Jewish holiday of Hanukkah. Each side has a letter of the Hebrew alphabet: נ (nun), ג (gimel), ה (hei), and ש (shin).



David presents a new game to his friends involving a dreidel and a fair coin. If the dreidel lands on ה (hei) and the coin lands on heads OR the dreidel lands on ג (gimel) and the coin lands on tails, the player wins.

- a. Define the sample space.

	נ (nun)	ג (gimel)	ה (hei)	ש (shin)
Heads	H נ	H ג	H ה	H ש
Tails	T נ	T ג	T ה	T ש

$\{H נ, H ג, H ה, H ש, T נ, T ג, T ה, T ש\}$

- b. How many outcomes result in a win?

2

- c. What is the probability a player wins?

$\frac{1}{4}$

**14.3.5 Wrap-Up: Reflection**

1. Color in the arrow up to the statement that best describes your current understanding of calculating theoretical probability for a two-step experiment.



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this in the next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

Answers will vary.

14.3.5 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons. Students need a concrete understanding of theoretical probability in order to compare it to experimental probability in the subsequent lessons.